

LWAZI KNOWLEDGE GUMEDE

ECSA Candidate Engineer | BSc (Honours) Computer Engineering | Rail Systems & Full-Stack Software

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PROFESSIONAL SUMMARY

ECSA Candidate Engineer building production software inside a live rail-infrastructure environment. Delivers full-stack monitoring systems in Python, Django, React, TypeScript and PostgreSQL that have replaced manual spreadsheet reporting for Transnet's Technical Office and corridor depots, and carries the same work through to field testing, commissioning and condition-assessment reviews. Works comfortably across the hardware and software boundary from an embedded and IoT background, and is currently reading for a master's in machine-learning fault detection for photovoltaic arrays.

WORK EXPERIENCE

Engineer-in-Training

April 2025 – Present

Transnet SOC Ltd (Rail Infrastructure Manager) | Johannesburg

- Designed, built and deployed the Rail Monitor signalling dashboard in React, TypeScript, Django, PostgreSQL and WebSocket, giving the Technical Office and corridor depots one live view of equipment availability across Points, Signals, TVDS, Power and Remote Control. Retired a manual spreadsheet cycle that had consumed close to a week of compilation every month.
- Built automated Python and REST services for data collection, vehicle identification and compliance reporting, removing recurring manual logging from daily engineering routines.
- Runs technical reviews, field testing and commissioning of Condition Assessment Systems (CAS), and supports derailment investigations with documented analysis and configuration management.
- Owns R&D engineering change proposals (ECPs) and the associated procurement workflows end to end across departments, from technical motivation through to award.

Engineering Tutor & Laboratory Demonstrator

February 2024 – May 2025

University of KwaZulu-Natal | 5 modules, first- and second-year engineering

- Ran Technical Communication (ENCH1TC) tutorials and gave structured written feedback on student reports and presentations, rated consistently well by lecturers and students for clarity.
- Demonstrated Computer Methods, Electronic Design, Electrical Principles (ENEL2EB) and Field Theory (ENEL2FT) laboratories, preparing and verifying practical equipment and supervising second-year cohorts for accurate results and full safety compliance.

Vacation Student (Engineering Internship)

January 2023 – March 2023

Transnet Engineering | Durban

- Built Power BI dashboards that turned raw engineering metrics into self-serve visual reports for stakeholder decision-making.
- Automated data hand-offs with Python pipelines that pushed engineering data into shared Google Sheets for real-time cross-department access.

EDUCATION

MSc, Computer & Electronic Engineering

Registering 2027

North-West University

Research area: machine-learning fault detection and localisation in photovoltaic arrays, using sparse neural network models over Simscape Electrical array simulations.

BSc (Honours) in Computer Engineering

2019 – June 2024

University of KwaZulu-Natal | ECSA-accredited

Focus areas: embedded systems, IoT, software development, digital signal processing, control systems, data structures and algorithms, database systems. Key projects: IoT smart meter system, blockchain invoice verification, multi-factor authentication system, PIC16 embedded game system, JPEG image compression.

National Senior Certificate (Matric)

2018

Nombuso High School | Bachelor's Degree Pass

TECHNICAL SKILLS

Languages: Python, C++, C, Java, JavaScript, TypeScript, PHP, SQL, Bash, MATLAB, Assembly

Web & Backend: Django, React, Next.js, Node.js, Express.js, Flask, RESTful APIs, WebSocket, HTML5, CSS3, Bootstrap

Data & Databases: PostgreSQL, MySQL, SQLite, MongoDB, database design and optimisation, Power BI, D3.js

Engineering Tools: MATLAB / Simulink, Proteus, Altium Designer, Arduino and microcontrollers, oscilloscope, Git, Docker, Linux, Postman, CI/CD

Domains: Rail condition monitoring and signalling, embedded and IoT systems, system automation, blockchain development, technical documentation, Agile/Scrum

Languages spoken: English (professional), isiZulu (native), isiXhosa (conversational)

PROFESSIONAL REGISTRATION

Candidate Engineer, Engineering Council of South Africa (ECSA) – registered 2025

Engineer-in-Training Programme – ECSA-accredited two-year structured training, Transnet SOC Ltd | April 2025 – March 2027. Competencies logged in a documented professional-development logbook. References available on request.